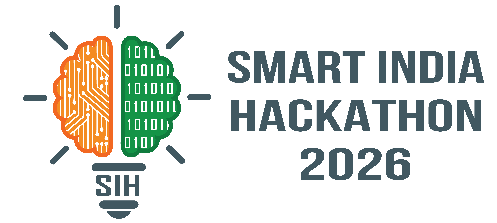


SMART INDIA HACKATHON 2026



RailSync

**AUTOMATIC BLOCK PLANNING
WITH CONTROLLED RECOVERY**

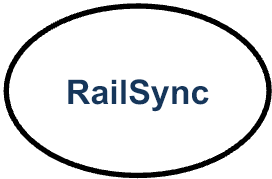
SIH26027 · Transportation & Logistics · Software

**AI-Powered Automatic Block Planning to Maximize Asset
Availability for Train Operations on Indian Railways**

TEAM NSUT082 · NETAJI SUBHAS UNIVERSITY OF TECHNOLOGY (NSUT)

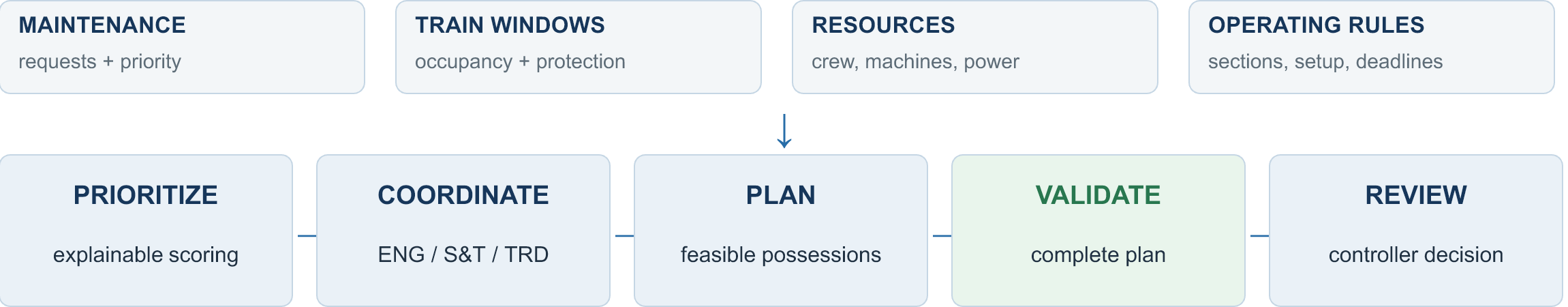
Prakhar Tyagi · Somya Sharma · Kunal Kumar
Priyanshu Chauhan · Gauransh · Shreyaan Awasthi





IDEA TITLE

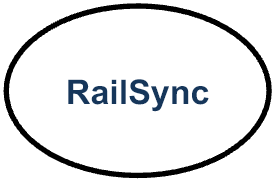
One workflow from maintenance demand to a reviewable block plan



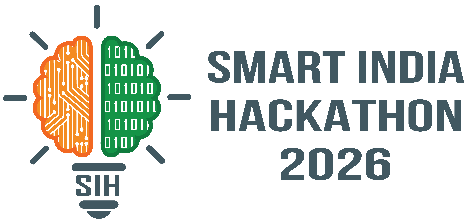
WHEN EXECUTION DEVIATES

Execution snapshot → resource-linked repair → complete-plan revalidation

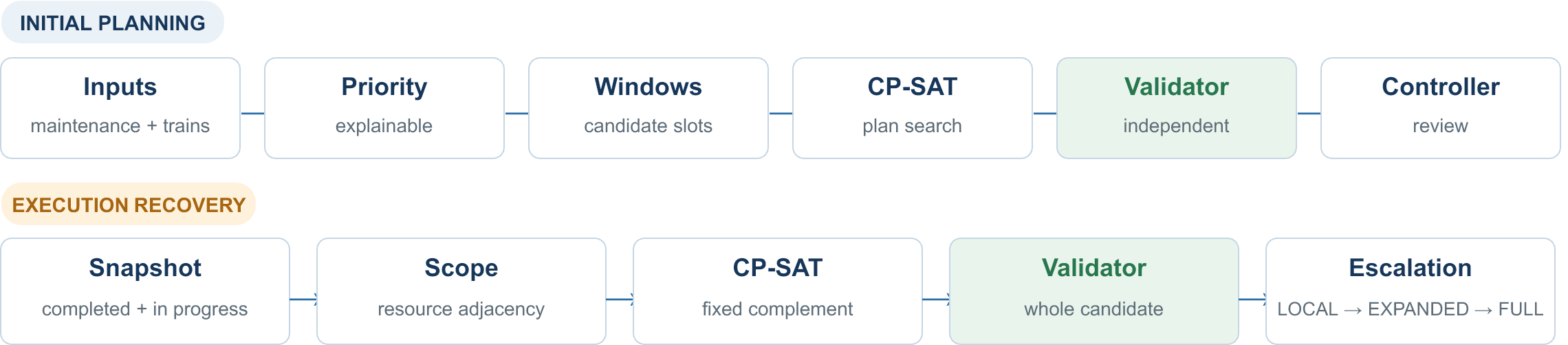
Unaffected approved commitments stay fixed; scope expands only when dependencies require it.



TECHNICAL APPROACH

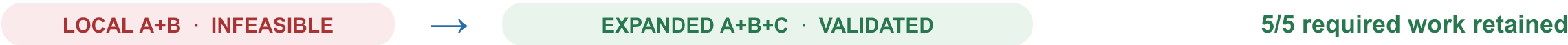


A single constraint model; a separate complete-plan validator

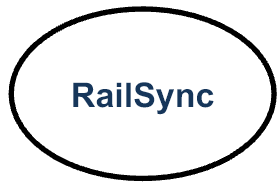


SYNTHETIC GOLDEN RECOVERY CASE 2

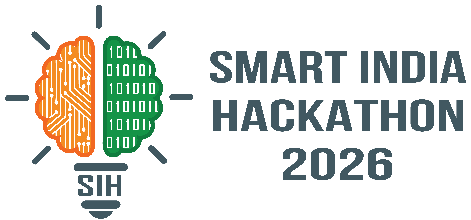
P_A — M_SHARED — P_B — PROTECTION_R — P_C



Python · OR-Tools CP-SAT · FastAPI/Pydantic · React/Vite · pytest



FEASIBILITY AND VIABILITY



Prototype evidence and the path to an authorized pilot

IMPLEMENTED & VERIFIED

408 automated tests pass
11 / 11 golden recovery tests pass
3 / 3 fixture contracts verified

Independent validation · execution-aware recovery
Stale proposal rejection · same CP-SAT model for FULL and RESTRICTED

DEPLOYMENT GAPS

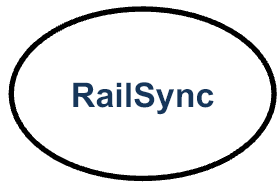
Authorized operational feeds and pilot rules
Production persistence and access control
Railway domain-expert validation
Corridor-scale benchmarking

SYNTHETIC CASE 3

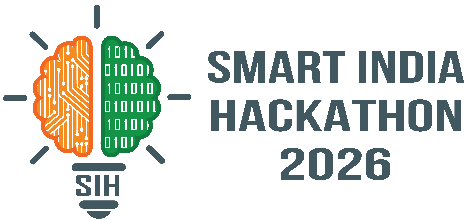
LOCAL → INFEASIBLE EXPANDED → INFEASIBLE FULL TERRITORY → INFEASIBLE

No adoptable repair; required work is not silently dropped.

Small synthetic fixtures validate behavior; they do not establish a general runtime advantage.



IMPACT AND BENEFITS



Operational value that the prototype can actually demonstrate

PLAN BETTER

Explainable priority
ENG / S&T / TRD coordination
Hard train and resource constraints

CHANGE LESS

Resource-linked scope
Preserve unaffected approved work
Escalate only when necessary

FAIL SAFELY

Independent complete-plan validation
No invalid candidate adoption
Controller remains in control

THREE DIFFERENT SYNTHETIC RECOVERY OUTCOMES

01 · LOCAL SUCCESS

P_A/P_B repaired; nearby independent P_U unchanged.

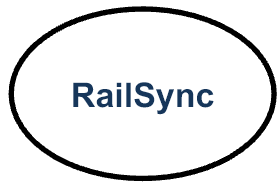
02 · BOUNDED EXPANSION

LOCAL infeasible → EXPANDED validated;
5/5 work retained.

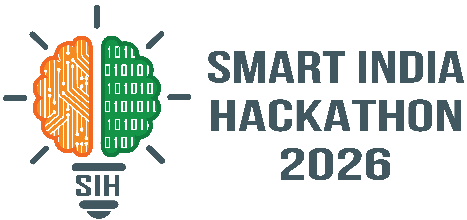
03 · HONEST INFEASIBILITY

No feasible service-floor repair; no candidate adopted.

Synthetic golden recovery fixtures · 408 automated tests passed across the current prototype



RESEARCH AND REFERENCES



Research grounding and inspectable implementation

PROBLEM CONTEXT

Ministry of Railways / Smart India Hackathon · SIH26027

SCOPE RESTRICTION

Nygren, Eichenberger & Frejinger (2023) · arXiv:2305.03574

MAINTENANCE DISRUPTION

Albrecht, Panton & Lee (2013) · DOI: 10.1016/j.cor.2010.09.001

WINDOW RESCHEDULING

Yang et al. (2026) · DOI: 10.1016/j.trc.2026.105781

OPTIMIZATION TOOL

Google OR-Tools · CP-SAT documentation

CODE github.com/prakhar9642/RailSync

PROTOTYPE rail-sync-eosin.vercel.app